

Boat Rental Company

Milestone 2

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Group 05 | 052400-1 VU Information Management and Systems Engineering

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What the system does

A boat rental company that operates out of several harbours.

- Clients search a harbour and a date range, book a boat and pay for the charter
- Managers run the fleet, the harbours, the staff and the bookings
- Both analytics reports are built into the application, and both run against MariaDB *and* MongoDB

Student 1, Ilja Golovanov

Book a rental, availability report

Student 2, Yernaz Togizbayev

Hire a staff member, supervision report

Infrastructure

Three containers, one command.

- **db** – MariaDB 11.3, the system of record
- **mongo** – MongoDB 7, the document database
- **backend** – Flask, waits for both to be healthy

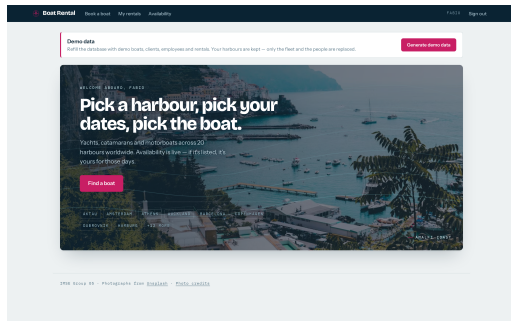
```
docker compose up --build # then http://localhost:5000
```

- Every dependency installed at image build, nothing needed on the host
- All three ports bound to 127.0.0.1: a development server and two databases with known credentials do not belong on a shared network

Data import

Generate demo data, on every page, no login needed.

- 200 boats over every harbour, 10 clients, employees, roughly 330 rentals
- Boats are generated *coherently*: length drives seats, power, weight, cabins and price
- Harbours are never deleted, or the only restocking tool would destroy every city a manager added
- One transaction, so a failure cannot leave it half-empty



Student 1

Ilja Golovanov

Use case: book a rental | Report: what is free, and where

Use case: book a rental

The client picks a harbour and dates, then picks a boat.

Boat Rental Book a boat My rentals Availability FABIO Sign out

23 SEP - 27 SEP 2026 · 4 DAYS

Boats in Auckland

10 of 10 free for these dates



Azimut 73 YACHT

10 GUESTS 22.4 METRES

Named "Sirocco" - no jacuzzi

€2,560.00 / day **€10,240.00**

B48 Auckland, New Zealand



Bayliner 18 MOTORBOAT

4 GUESTS 5.5 METRES

Diesel outboard engine

€270.00 / day **€1,680.00**

B108 Auckland, New Zealand



Bayliner 33 MOTORBOAT

10 GUESTS 10.1 METRES

Diesel inboard engine

€690.00 / day **€2,760.00**

B68 Auckland, New Zealand



Boston Whaler 34 MOTORBOAT

9 GUESTS 10.3 METRES

€1,200.00 / day **€4,800.00**

B108 Auckland, New Zealand



Boston Whaler 30 MOTORBOAT

9 GUESTS 9.1 METRES

€1,000.00 / day **€4,000.00**

B108 Auckland, New Zealand



Lagoon 46 CATAMARAN

9 GUESTS 14.1 METRES

€1,800.00 / day **€7,200.00**

B108 Auckland, New Zealand

The whole fleet is listed. Booked and out-of-service boats are greyed, labelled, and carry no radio at all, so they cannot be selected.

Booking creates a hold, not a charter

[Boat Rental](#) [Book a boat](#) [My rentals](#) [Availability](#) [FABIO](#) [Sign out](#)

Demo data
Refill the database with demo boats, clients, employees and rentals. Your harbours are kept — only the fleet and the people are replaced.
[Generate demo data](#)

CHECKOUT

Pay for your charter
Your boat is reserved. It is confirmed once this is paid.

Booking summary

BOAT	B161 • Quicksilver
HARBOUR	Mykonos, Greece
FROM	12 Sep 2026
TO	16 Sep 2026
DAILY RATE	€410.00 × 4 days
Total	€1,640.00

Demo payment. No real card is charged and no card details are stored.

- Pay: 4242 4242 4242 4242
- See a decline: 4000 0000 0000 0002

Any future expiry date and any 3-digit CVC will do.

NAME ON CARD

CARD NUMBER

EXPIRY

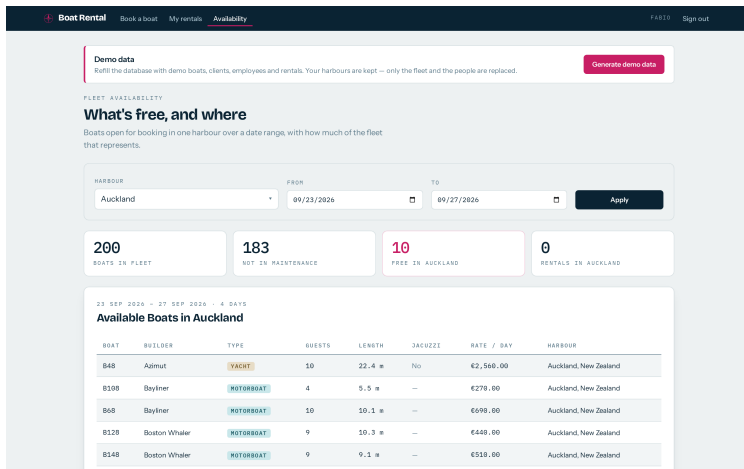
CVC

[Pay now](#) [Pay later](#)

You can pay later, but the boat is only held until **11 Sep, 09:00** — 24 hours before your charter starts. Unpaid after that, the reservation is cancelled.

An unpaid booking holds the boat until 24 hours before the ride. Book inside that window and the hold is 15 minutes, counted down live. When it lapses the boat goes back on the market.

Analytics report: availability



Free boats in one harbour over a date range, against the fleet size. Booking a boat removes it from this report: the page and the booking form share one overlap rule, so they cannot disagree.

Student 2

Yernaz Togizbayev

Use case: hire a staff member | Report: who supervises whom

Use case: hire a staff member

+

Boat Rental

Employees

Cities

Fleet

Rentals

Supervision

Maintenance

MongoDB

MANAGER · MORGAN HOORE

Switch

Sign out

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Generate demo data

ROSTER

Hire an employee

OFFICE

Aktau (Ofa3ec8c4)

FIRST NAME

LAST NAME

STREET

ZIP

CITY

COUNTRY

BIRTHDATE

mm/dd/yyyy

EMAIL

MOBILE NUMBER

SELF INSURANCE NUMBER

SALARY

ROLE

Staff

WORK SHIFT

Day

☐ Is on duty?

Hire

Cancel

Exercises the IS-A hierarchy: one **Employee** row plus a **Staff** or **Manager** row. Changing someone's role later moves them between the two subclass tables.

Analytics report: supervision

Boat Rental

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MANAGER · MORGAN MOORE

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Generate demo data

REPORTING LINES

Supervision

Which manager supervises which staff member.

MANAGER

STAFF MEMBER

Assign supervision

MANAGER	STAFF MEMBER	ACTIONS
Alex Brown (E8)	Jamie Davis (E2)	Unassign
Alex Brown (E8)	Riley Smith (E6)	Unassign
Morgan Moore (E7)	Jamie Anderson (E4)	Unassign
Morgan Moore (E7)	Riley Anderson (E1)	Unassign
Morgan Moore (E7)	Avery Miller (E5)	Unassign
Morgan Moore (E7)	Quinn Thomas (E3)	Unassign

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Which staff a manager supervises, with shift, duty status and office. Hiring a staff member and assigning them adds a row here: three became four.

NoSQL

MongoDB

Re-design, migration, both reports, indexing

Re-design: shaped around the questions

Ten tables become three collections. Each use case's answer lives in one document.

Collection	Embeds	Serves
offices	boats, and each boat's rentals	Student 1's report
managers	the staff they supervise	Student 2's report
clients	their own rentals	the client logbook

- Primary keys kept as `_id`; they are meaningful and already in the URLs
- Most foreign keys disappear: `Supervises` stops being a table, the relationship *is* the array
- IS-A flattened, since nothing asks for "an employee" without knowing which kind

Re-design: the trade

```
{ "_id": "05", "city": "Dubrovnik", "country": "Croatia",  
  "boats": [ { "boatId": "B27", "type": "catamaran", "dailyRate": "1120.00",  
    "rentals": [ { "clientId": "C3", "rentalDate": "2026-09-04",  
      "paymentStatus": "PAID" } ] } ] }
```

- A rental is stored *twice*, under its boat and under its client
- Both reads become single-document; the duplication costs nothing at runtime because only the migration writes
- It would be the first thing to revisit if the app wrote to MongoDB directly

Migration

[Boat Rental](#) [Employees](#) [Cities](#) [Fleet](#) [Rentals](#) [Supervision](#) [Maintenance](#) [MongoDB](#) [MANAGER · MORGAN MOORE](#) [Switch](#) [Sign out](#)

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MONGODB

The same questions, asked of documents

Both analytics reports run here against the document model instead of the relational one. The answers should match what the SQL pages show.

Migration

Clears every collection and rebuilds it from MariaDB. One-way: the relational database stays the system of record.

[Migrate to MongoDB](#)

20
OFFICES

3
MANAGERS

10
CLIENTS

[Index statistics](#)

STUDENT 1 · GOLOVANDY

Boats free in a harbour

One aggregation over **offices**. No joins: the boats and their rentals are inside the harbour document.

HARBOUR

FROM

TO

Aktau

08/24/2026

08/31/2026

Run

3 free in Aktau.

BOAT

DEFINED

TYPE

REQUEST

LENGTH

TAPOUT

DATE / DAY

MANAGER

One button, clears the collections first so a second run changes nothing. One-way: MariaDB stays the system of record. Reads through the existing models, so there is one definition of what a boat is.

Both reports, on documents

Student 1 – one aggregation

```
db.offices.aggregate([
  { $match: { city: "Dubrovnik" } },
  { $unwind: "$boats" },
  { $match: { "boats.availabilityStatus":
    "Available" } },
  ...filter overlapping rentals...
])
```

Student 2 – one document

```
db.managers.find({ _id: "E7" })
```

Replaces a five-way join across Supervises, Staff, Employee twice, Manager and Office.

Verified, not assumed: all 20 harbours return the same free boats from SQL and MongoDB, and every manager the same staff.

Indexing

[Boat Rental](#) [Employees](#) [Cities](#) [Fleet](#) [Rentals](#) [Supervision](#) [Maintenance](#) [MongoDB](#) MANAGER · MORISAN MOORE [Switch](#) [Sign out](#)

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MONGODB

Index statistics

Each report's query run twice through `explain("executionStats")`, once with its index dropped and once with it in place.

OFFICES · CITY_IDX

Boats free in Aktau

INDEX	DOCS EXAMINED	KEYS EXAMINED	RETURNED	PLAN	INDEX USED
dropped	20	0	1	PROJECTION_SIMPLE	—
in place	1	1	1	PROJECTION_SIMPLE	city_idx

MANAGERS · SUPERVISED_STAFF_IDX

Which manager supervises E1

INDEX	DOCS EXAMINED	KEYS EXAMINED	RETURNED	PLAN	INDEX USED
dropped	3	0	1	COLLSCAN	—
in place	1	1	1	FETCH	supervised_staff_idx

Both indexes are back in place; dropping them is only done for the measurement above.

offices.city for Student 1: 20 documents examined down to 1.

managers.supervisedStaff.employeeId for the reverse of Student 2's report: COLLSCAN to IXSCAN. Small in absolute terms at this data size; what changes is that the scan grows with the collection and the lookup does not.

Summary

- Three containers, one command, nothing installed on the host
- Both use cases and both analytics reports implemented twice, on MariaDB and on MongoDB, and cross-checked against each other
- One-way migration behind a button, collections cleared first
- Indexes measured with `explain`, not assumed

Questions?